Validity Analysis: CRISISLTLSUM

Target context: Mexico City and Central America Bilingual Crisis Journalism

Overall risk: **HIGH**

Deployment Context

Use case: Summarization of quick messages (e.g. tweets) for event reporting in the crisis domain for news writing in American English and Mexican Spanish mostly

Target population: Journalists and media workers who monitor social media to cover breaking news from Mexico City. They would speak Mexican Spanish natively (possible neutral/standard regional variety), and English fluently. They would be expected to use the model in English and Spanish to cover news of disasters happening not only in Mexico, but also south of the United States and also Central America.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ⚠	1	Serious concern	high
Input Content ⚠	1	Serious concern	high
Input Form ✓	3	Moderate gaps	medium
Output Ontology	2	Significant gaps	high
Output Content ⚠	1	Serious concern	high
Output Form ⚠	1	Serious concern	high
Average	**1.5**		

⚠ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

CrisisLTLSum is a fundamentally English-language, US/Australian-anchored crisis-tweet benchmark whose four-domain ontology (wildfire/local fire/traffic/storm) [Q1, Q122], English-keyword-filtered content [Q126], English NER + OpenStreetMap pipeline [Q35, Q36], Anglophone MTurk annotators [Q61], English-anchored coherence rubric [Q153], and English-only ROUGE evaluation [Q99, Q100] together create severe validity gaps for the target deployment of bilingual crisis journalism in Mexico City and Central America. The structurally important crisis types for the deployment area — earthquakes, volcanic activity (Popocatépetl), tropical-system flooding, civil unrest, and CDMX transportation infrastructure failures — are entirely absent from the benchmark [WEB-7, WEB-20, WEB-23]. The Spanish-language output requirement (one of two deployment languages) receives zero coverage from the benchmark's evaluation framework. Of the six dimensions, five are scored 1–2 (high concern) and only Input Form scores 3 due to modality match. The authors themselves acknowledge non-portability to other languages and regions [Q119].

ID	Evidence
Q1	“CrisisLTLSum contains 1,000 crisis event timelines across four domains: wildfires, local fires, traffic, and storms.” (p.1)
Q122	“We build CrisisLTLSum by limiting the search to four crisis domains: wildfire, local fire, traffic, and storm.” (p.12)
Q126	“We carefully curate a keyword list for each of our categories of interest by employing expert knowledge gathered through over-viewing crisis stories in news, social media, and related big disaster events of the past.” (p.13)
Q35	“Since location mentions are crucial in extracting local events and existing models have low performance detecting them from noisy tweets text, we further developed a BERT-based NER model tuned on Twitter data to detect location mentions.” (p.4)
Q36	“Location Augmentation We use OpenStreetMap API to map location mentions to physical addresses.” (p.4)
Q61	“First, we use location restriction (QC1) to limit the pool of workers to countries where native English speakers are most likely to be found.” (p.5)
Q153	“For each summary, answer the question "how coherent is the summary on its own?" on a scale 1 to 5. A summary is coherent if when read by itself, it's easy to understand and free of English errors. A summary is not coherent if it's difficult to understand what the summary is trying to say. Generally, it's more important that the summary is understandable than it being free of grammar errors.” (p.21)
Q99	“The summarization output of each model is evaluated in a multi-reference setting against the two summaries written by the two workers who agree the most based on the timeline extraction labels.” (p.8)
Q100	“We use SacreROUGE (Deutsch and Roth, 2020) to compute multi-reference ROUGE scores.” (p.8)
Q119	“Accordingly, replicating the same process for other language or based on other locations may be hard because of such dependencies.” (p.10)
WEB-7	SASMEX vocabulary defines a specific seed-tweet/follow-up structure not addressed by benchmark output categories (https://en.wikipedia.org/wiki/Mexican_Seismic_Alert_System)
WEB-20	CENAPRED semáforo phase vocabulary is essential output-ontology content for volcanic crises (https://www.cenapred.unam.mx/reportesVolcanesMX/)
WEB-23	#Verificado19S citizen-verification genre would be judged differently by Mexican journalists than US MTurk workers (https://arxiv.org/abs/2007.05848)

Practical Guidance

What This Benchmark Measures

CrisisLTLSum measures, for English-language US/Australian tweets in four crisis domains (wildfire, local fire, traffic, storm), how well systems can (a) extract relevant non-redundant updates from a noisy tweet stream given a seed tweet [Q5, Q83] and (b) produce abstractive summaries scored against multi-reference human summaries [Q98, Q99]. Within that English/US scope, it provides defensible signal: 90%+ inter-annotator agreement [Q77] and a 73.51% vs. 88.98% model-vs-human gap [Q96] indicate the tasks are non-trivial and well-instrumented. For the target deployment, the benchmark's English-output evaluation framework can serve as a starting point for the English half of the bilingual requirement.

Construct Depth

Construct depth is shallow for the target deployment. The benchmark probes timeline extraction and summarization only on English tweets in four event types with no Spanish content, no journalist annotators, no editorial-style axes, no semáforo or SASMEX vocabulary, and no register-appropriateness metric. Even where the benchmark provides defensible signal (English summarization quality on fire/traffic/storm events), it does not address whether systems handle Mexican Spanish, voseo, code-switching, alcaldía/colonia toponymy, CENAPRED/SSN attribution, the #Verificado19S verification-tweet genre, or AP Español register conformance. The Coherence rubric [Q153] is structurally inapplicable to Spanish output. Strongest dimension is Input Form (modality match); weakest are Input Ontology, Input Content, Output Content, and Output Form — all of which would need substantial supplementation.

What Else You Need

A complete assessment requires: (1) construction of a Spanish-language crisis tweet corpus anchored to Mexican/Central American events covering earthquakes, volcanic activity, tropical flooding, and civil unrest (no such corpus currently exists per [WEB-3, WEB-4]); (2) re-annotation by Mexican journalist annotators using an extended relevance/informativeness rubric that accounts for verification posts [WEB-23] and community solidarity posts; (3) a Spanish coherence rubric and AP Español–aligned register/attribution axes to replace [Q153]; (4) a bilingual summarization evaluation protocol using BERTScore-multilingual and LLM-as-judge alongside ROUGE [WEB-17, WEB-18]; (5) Spanish gazetteer integration for alcaldía/colonia/municipio resolution to replace the OSM-English pipeline [Q36]; and (6) explicit handling of SASMEX/CENAPRED hashtag and semáforo vocabulary as seed-tweet types [WEB-7, WEB-20].

ID	Evidence
Q5	“The timeline extraction task is formalized as: given a seed tweet as the initial mention of a crisis event, extract relevant tweets with updates on the same event from the incoming noisy tweet stream.” (p.1)
Q83	“Given an initial seed tweet t0, and following chronologically sorted tweets ti, i = 1, ..., n, the goal of the timeline extraction task is to determine whether tweet ti is part of the timeline, i.e., related to the same event and adding new information compared to all prior tweets [t0, ..., ti−1].” (p.6)
Q98	“We evaluate all timeline summarization models with ROUGE (Lin and Och, 2004).” (p.8)
Q99	“The summarization output of each model is evaluated in a multi-reference setting against the two summaries written by the two workers who agree the most based on the timeline extraction labels.” (p.8)
Q77	“The average timeline-level agreement between those two workers is 90.06%.” (p.6)
Q96	“The best timeline extraction model achieves 73.51 average timeline accuracy, whereas the human-level performance is 88.98.” (p.8)
Q153	“For each summary, answer the question "how coherent is the summary on its own?" on a scale 1 to 5. A summary is coherent if when read by itself, it's easy to understand and free of English errors. A summary is not coherent if it's difficult to understand what the summary is trying to say. Generally, it's more important that the summary is understandable than it being free of grammar errors.” (p.21)
Q36	“Location Augmentation We use OpenStreetMap API to map location mentions to physical addresses.” (p.4)
WEB-3	HumAID is English-only despite including a 2017 Mexico earthquake subset — confirming the gap in Spanish-language Mexican crisis tweet content (https://crisisnlp.qcri.org/humaid_dataset.html)
WEB-4	Curiel et al. (PMC 2019) is the only known Mexican-Spanish crisis NER/geolocation resource and is not a timeline/summarization dataset (https://pmc.ncbi.nlm.nih.gov/articles/PMC6484392/)
WEB-23	#Verificado19S citizen-verification genre would be judged differently by Mexican journalists than US MTurk workers (https://arxiv.org/abs/2007.05848)
WEB-17	Standard ROUGE variants correlate weakly with human judgments for Spanish summarization; BERTScore-precision with multilingual models is recommended (https://arxiv.org/pdf/2503.17039)
WEB-18	LLM-as-judge approaches show strong correlation with human judgments in multilingual summarization evaluation (https://aclanthology.org/2024.emnlp-main.1085.pdf)
WEB-7	SASMEX vocabulary defines a specific seed-tweet/follow-up structure not addressed by benchmark output categories (https://en.wikipedia.org/wiki/Mexican_Seismic_Alert_System)
WEB-20	CENAPRED semáforo phase vocabulary is essential output-ontology content for volcanic crises (https://www.cenapred.unam.mx/reportesVolcanesMX/)